ChronoLens: Intelligent Video Retrieval and Temporal Reasoning for Large-Scale Video Analysis

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***Abstract*—Finding a specific moment in a large video archive often depends on several kinds of evidence at once: what is visible, which objects persist over time, what is said, and what text appears on screen. This paper presents ChronoLens, a multimodal retrieval framework that turns raw video into a reusable evidence index and answers natural-language queries by routing each query to the evidence it needs and then verifying candidates against tracked entities. At indexing time, YOLO26m and ByteTrack produce persistent object tracks, OpenCLIP ViT-L/14 embeddings are stored in FAISS for candidate retrieval, and Whisper and PaddleOCR supply timestamped speech and scene text. At query time, attribute-constrained queries are checked against individual tracks and relational queries against pairs of tracks, and the surviving candidates are ranked by a score that combines semantic similarity, detector confidence, and track persistence. Every result points back to its source frame, track, and timestamp. Five cumulative configurations were compared on a 107-query benchmark built on a single video. The full system reaches 68.2% R@1, 91.6% R@10, 0.75 MRR, and 73.8% mAP@10, and lowers the negative-control false-positive rate from 21.0% to 8.0%, at a mean query latency of 2.81 s.**

***Index Terms*—video retrieval, multimodal retrieval, semantic search, object tracking, temporal reasoning, computer vision, CLIP, YOLO, ByteTrack, FAISS**

I. Introduction

Cameras in public spaces, on phones, and on web platforms produce recordings far faster than people can watch them, so the ability to search video by description has become a practical necessity. Many existing tools depend on manually entered tags or on frame-by-frame visual similarity. Neither handles a question such as "the person in a yellow shirt who walks past a bicycle before someone speaks," which needs the person's appearance, another object, the order of events, and possibly audio or on-screen text to be considered together. Retrieval of this kind is closer to multimodal reasoning than to image matching.

Re-running detection, tracking, speech recognition, and text recognition for every query is wasteful. A cheaper design analyzes the video once, stores the results as an index, and answers many queries from that index. Tevissen et al. [1] follow this idea for video libraries, where a language model writes search queries over indexed speech and visual metadata and the system returns timestamped moments. Dela Rosa [2] instead converts video into aligned textual captions so that text-based retrieval and language models can work with visual content. Both works suggest keeping costly analysis separate from query handling.

Similarity in an embedding space, however, does not always mean that all the conditions of a query are satisfied. For example, a frame may receive a high similarity score for "person wearing yellow shirt" because a yellow object is close to a person, even when the person is not wearing yellow. Similarly, "person near bicycle" describes a spatial relationship between two entities. It cannot be answered reliably by finding one frame containing a person and another frame containing a bicycle. ChronoLens therefore uses semantic search mainly to generate possible candidates and then checks the explicit query conditions against structured evidence.

ChronoLens follows this approach by first decoding video into representative frames. YOLO26m detects objects and ByteTrack connects detections into persistent tracks. Attribute V3 summarizes the appearance of each person track over multiple frames. OpenCLIP ViT-L/14 generates embeddings for frames and object crops, which are indexed in FAISS. Whisper converts speech into timestamped text, while PaddleOCR extracts text visible in the scene. A structured store records the connections between videos, frames, detections, tracks, attributes, speech, and scene text.

At query time, a router identifies the main type of evidence required by the query. Object queries use detector output, attribute queries use track-level attributes, and relational queries use pairs of tracks. Temporal, speech, and OCR queries use their corresponding timestamped records. Other queries fall back to semantic retrieval. Candidates that do not satisfy the stated conditions are removed, and the remaining results are ranked using semantic similarity, detector confidence, and track persistence.

The main contributions of this work are:

- an evidence index connecting frames, detections, persistent tracks, attributes, speech, scene text, and timestamps;
- a query router that directs natural-language queries toward the evidence source required to answer them;
- an attribute verification step that requires all requested attributes to belong to the same persistent track;
- a relational verification step that checks pairs of tracked entities in spatial and temporal context;
- a ranking method that combines semantic similarity, detector confidence, and track persistence while keeping each result linked to its source evidence.

The remainder of the paper is organized as follows. Section II discusses related work. Section III describes the ChronoLens architecture. Section IV explains retrieval and query routing. Section V presents the experimental methodology, Section VI discusses the results, and Section VII concludes the paper.

II. Related Work

A. Video-Text Retrieval and Moment Retrieval

CLIP [3] learns a shared representation for images and text, allowing a natural-language phrase to be compared with visual content. CLIP4Clip [4] investigates how this representation can be applied to video retrieval and captioning. Moment-DETR [5] extends natural-language video retrieval by identifying the relevant time span within a video. These methods are useful for understanding the semantic meaning of a query, but they do not directly verify whether a particular tracked entity satisfies every condition expressed in that query.

B. Retrieval-Augmented Video Understanding

Tevissen et al. [1] use retrieval-augmented generation for video libraries. Their approach generates search queries, retrieves moments using speech and visual metadata, and produces answers with timestamps. Dela Rosa [2] represents video using aligned captions so that retrieval and language-model reasoning can operate over textual representations.

Both approaches analyze the video before the query is submitted, which is also an important design principle in ChronoLens. However, ChronoLens focuses on returning evidence records linked to frames, tracks, and timestamps rather than directly generating a final natural-language answer.

C. Detection, Tracking, Speech, and Vector Search

ByteTrack [6] associates detection boxes across frames to create object trajectories and persistent identities. DeepSORT [7] uses learned appearance information for data association and tracking. Whisper [8] provides timestamped speech transcription, allowing spoken content to be treated as time-aligned evidence. FAISS [9] provides efficient similarity search over dense vectors.

The YOLO family [10] provides real-time object detection, while YOLO26 [11] is the recent model used in this work. OpenCLIP representations are used for the semantic retrieval component [14], and PaddleOCR provides scene-text evidence [15]. These components serve different purposes in ChronoLens. Vector search is used to propose candidates, while tracks, transcripts, detector confidence, and structured metadata provide the evidence needed for later verification.

D. Person Search and Positioning of ChronoLens

Searching for people using natural-language descriptions has been studied in person search [12] and person re-identification [13]. Such approaches mainly focus on matching people across images. ChronoLens instead stores person-level attributes as part of persistent tracks in an evidence index that also contains objects, speech, and scene text.

The main idea is to keep frames, detections, tracks, attributes, timestamps, speech, and scene text connected so that query conditions can be checked after candidate retrieval. This also allows multiple attributes to be required for the same track and allows relationships to be examined between two tracks using their spatial and temporal context.

III. ChronoLens System Architecture

ChronoLens contains two connected stages, as shown in Fig. 1. The first stage processes the raw video and creates reusable evidence in a vector store and a structured store. The second stage interprets a query, retrieves possible candidates, checks the query conditions, ranks the remaining candidates, and returns the top-K results.

Since the expensive video analysis is performed during indexing, repeated queries do not require the original video to be processed again. Each returned result can also be traced back to a location in the source video.

A. Storage and Indexing Pipeline

The uploaded video is first decoded using FFmpeg, while OpenCV is used to extract representative frames. Each frame is stored with its frame index and timestamp. YOLO26m detects objects in the frames, and each detection contains its class, bounding box, timestamp, confidence, and track identity. ByteTrack associates detections across consecutive frames, allowing objects to be followed over time.

For person tracks, Attribute V3 works with crops obtained from several frames rather than relying on a single image. Pooling information across multiple frames gives a more consistent description of the same person. The stored attributes include clothing, colour, accessories, hair, viewpoint, and activity.

The evidence associated with frame i is represented as

$$E_i = \{F_i, O_i, T_i, A_i, X_i, R_i\}. \tag{1}$$

Here, F_i represents the frame and its timestamp, O_i contains detected objects and their positions, T_i represents persistent tracks, A_i stores track-level attributes, X_i contains OCR and speech evidence, and R_i stores relation metadata.

For semantic retrieval, image and text embeddings are generated using OpenCLIP ViT-L/14 [14]. Whisper provides timestamped speech transcripts, while PaddleOCR provides text detected in frames. These textual sources remain associated with their corresponding times and can be searched separately or together with visual evidence.

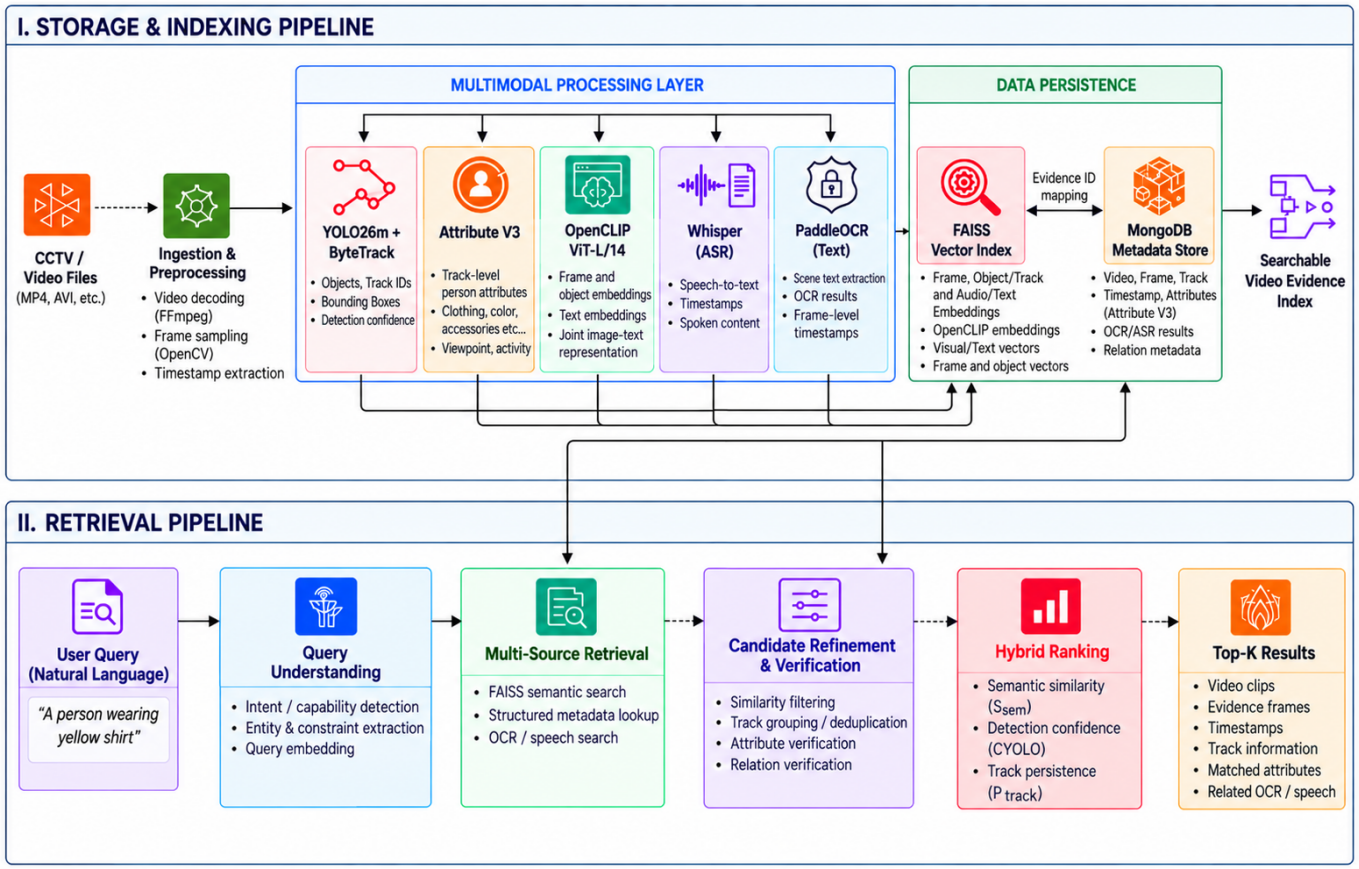


Fig. 1. ChronoLens architecture showing the Storage and Indexing Pipeline (top) and the Retrieval Pipeline (bottom).

B. Data Persistence Layer

ChronoLens uses two main storage components. FAISS stores the embeddings and performs nearest-neighbour search. MongoDB stores the structured records connecting videos, frames, detections, tracks, attributes, OCR segments, and speech segments.

The evidence hierarchy can be represented as

$$\text{Video} \rightarrow \text{Frame} \rightarrow \text{Detection} \rightarrow \text{Track}.$$

This structure allows a vector returned from FAISS to be resolved to its corresponding structured record and then to the timestamp in the original video.

C. Retrieval Pipeline

The first step during retrieval is to understand what type of evidence the query needs. Object and attribute queries use detection and track records, while relational queries operate on pairs of tracks. OCR and speech queries use their own timestamped text. Queries without a specialized condition use semantic retrieval.

Before ranking, the conditions expressed in the query are checked. Attribute queries require all requested attributes to belong to the same track, while relational queries require the specified relation to hold for the relevant pair of tracks. Candidates that pass these checks are then ranked using the hybrid score and the top-K results are returned.

IV. Retrieval Pipeline and Query Routing

Once the index exists, each query passes through five stages: query understanding and routing, semantic retrieval, candidate pooling, constraint verification, and hybrid ranking. Semantic retrieval provides coverage of possible matches, while structured evidence is used to check the conditions specified by the query.

A. Query Understanding and Capability Routing

Query handling is treated as choosing which capability to use. The router identifies the main requirement of the query and extracts the entities and constraints that must be verified later. Queries about common objects are matched to detector classes. Clothing and human-attribute queries use Attribute V3 records.

Relational queries operate on pairs of tracked entities, temporal queries use timing information, and OCR and speech queries read their text evidence. Any other query falls back to semantic retrieval.

For example, "person wearing yellow shirt" is a clothing-attribute query and is sent to person tracks and their attribute

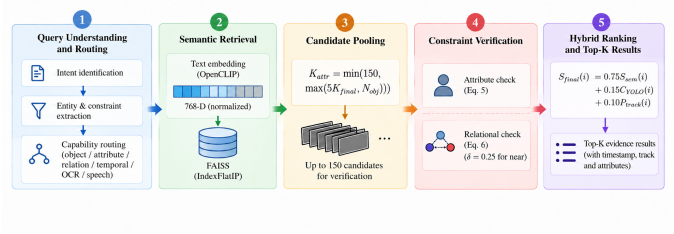


Fig. 2. Retrieval pipeline of ChronoLens: query understanding and routing, semantic retrieval, adaptive candidate pooling, constraint verification, and hybrid ranking produce evidence-linked Top-K results.

records. "Person near bicycle" is sent to relational processing, where both entities and their spatial relation are checked.

B. Adaptive Candidate Pooling

The number of candidates gathered is kept independent of the number of final results returned, so that verification has enough candidates to work with. For attribute-oriented retrieval, the candidate count is

$$K_{\text{attr}} = \min(150, \max(5K_{\text{final}}, N_{\text{obj}})). \tag{2}$$

Here, K_{final} represents the number of results requested by the user and N_{obj} represents the number of object vectors available in the index. The inner maximum aims to provide a candidate pool of at least five times the requested result count when sufficient object vectors are available. The outer minimum limits the pool to 150 candidates.

If no object vectors are available, the system uses K_{final} directly. Thresholding, grouping, attribute verification, and final ranking are then performed before the results are returned.

C. Semantic Candidate Retrieval

Semantic candidates are generated using OpenCLIP ViT-L/14 [14]. Let e_i denote an indexed embedding and e_q the query embedding. Both vectors are normalized to unit length. Their similarity is calculated as

$$S_{\text{sem}}(q, i) = e_q^{\top} e_i, \qquad \|e_q\|_2 = \|e_i\|_2 = 1. \tag{3}$$

Since both vectors have unit length, their inner product is equivalent to cosine similarity. FAISS retrieves the closest vectors, which are then resolved using the structured store. The `laion2b_s32b_b82k` checkpoint provides 768-dimensional normalized embeddings. FAISS IndexFlatIP indexes are used for both frame and object vectors.

D. Hybrid Ranking

After the constraints have been checked, the remaining candidates are scored using

$$S_{\text{final}}(i) = 0.75S_{\text{sem}}(i) + 0.15C_{\text{YOLO}}(i) + 0.10P_{\text{track}}(i). \tag{4}$$

Here, S_{sem} represents semantic similarity, C_{YOLO} represents detector confidence, and P_{track} represents track persistence. The weights were selected heuristically and kept fixed for all evaluated configurations.

Because ranking is performed after verification, a candidate that fails an attribute or relational condition cannot return to the final results simply because it has a high semantic similarity score.

E. Track-Level Attribute Verification

When an attribute-constrained query is received, its requested attributes are checked against one persistent track t:

$$\text{Match}_{\text{attr}}(q, t) = \prod_{a \in A(q)} \mathbf{1}[a_t = a_q]. \tag{5}$$

Here, $A(q)$ represents the set of attributes mentioned in query q, a_t is the value of attribute a recorded for track t, and a_q is the value requested by the query.

The product becomes 1 only when all requested attributes match on the same track. Attribute V3 uses up to 10 representative frames for each track. Clothing and human-attribute queries are also restricted to the person class when the query requires a person.

F. Spatial-Temporal Relational Verification

Relations are represented using a graph $G = (V, E)$, where the nodes represent tracked entities and the edges represent relationships between them. For a query involving two tracks t_i and t_j, verification is defined as

$$\text{Match}_{\text{rel}}(q, t_i, t_j) = \mathbf{1}[d(i, j) \le \delta]\, \mathbf{1}[T(i, j) \models q]. \tag{6}$$

Here, $d(i, j)$ is the Euclidean distance between the bounding-box centres normalized by the frame diagonal, δ is the spatial threshold, and $T(i, j)$ represents the temporal relation required by the query.

For "near," $\delta = 0.25$. Since the distance is normalized, the threshold has no physical unit. For "riding," the threshold is 0.30 and is combined with a bounding-box overlap test. Spatial relationships are evaluated using detections appearing together in the same frame. Temporal relationships such as "before" and "after" are determined from the start times of the corresponding tracks.

Candidates that fail the relational verification step are removed before ranking.

G. Evidence-Linked Top-K Retrieval

After candidate generation, filtering, verification, and ranking, the top-K evidence is returned. Each result remains connected to its source frame, timestamp, track, and any supporting attribute, OCR, or speech record. This provides a traceable connection between the query and the corresponding location in the original video.

V. Experimental Methodology

A. Evaluation Setup

ChronoLens was evaluated using a fixed set of 107 queries covering semantic, object, attribute, relational, temporal, speech, and OCR retrieval. The test video contains 8,853 frames at 23.976 fps and is approximately six minutes long.

Keyframes were extracted at one-second intervals, resulting in 369 analyzed keyframes.

The resulting index contains 251 person tracks, 72 timestamped speech segments, 369 frame embeddings, and 753 object embeddings.

Detection and tracking use YOLO26m and ByteTrack. Semantic retrieval uses OpenCLIP ViT-L/14, text extraction uses PaddleOCR, speech transcription uses Whisper, and vector search uses FAISS. A structured store is used to associate the retrieved evidence with its source information.

The experiments were conducted on an NVIDIA GeForce RTX 4050 with approximately 6 GB of memory using PyTorch 2.2.0+cu121 and CUDA 12.1. OpenCLIP and YOLO26m were executed on the GPU, while FAISS was executed in host memory.

Query latency was measured from query submission until the final top-K list was returned. The measurement includes both retrieval and ranking.

B. Retrieval Configurations

Five cumulative configurations were evaluated, with each configuration adding one capability to the previous one. The source video, benchmark, ground-truth procedure, and evaluation metrics remained the same for all configurations.

TABLE I
CUMULATIVE RETRIEVAL CONFIGURATIONS

Configuration	Capabilities	Description
B1: Frame-Semantic	Semantic	Frame-level ranking by CLIP text similarity.
B2: Object-Aware	Semantic + Object	Adds object retrieval and track-linked evidence.
B3: Attribute-Aware	Semantic + Object + Attribute	Adds track-level attribute verification with Attribute V3.
B4: Relational	Semantic + Object + Attribute + Relation	Adds spatial-temporal relational reasoning.
B5: Full ChronoLens	Complete pipeline	Adds speech, OCR, and evidence linking to B4 components.

C. Query Benchmark and Ground Truth

The benchmark contains 107 queries divided into nine categories, as shown in Table II.

TABLE II
107-QUERY EVALUATION BENCHMARK

Query Category	Number of Queries
Clothing Attributes	20
Human Attributes	15
Multi-Attribute	15
Objects	15
Relations	15
Negative Controls	10
Temporal	10
Speech	4
OCR	3
Total	**107**

Relevance was determined by inspecting the source video together with ChronoLens evidence and metadata. A result was considered relevant when its evidence satisfied the query, regardless of its semantic score. The same criterion was applied to B1–B5.

Because ChronoLens evidence contributed to the relevance judgements, the ground truth is not completely independent of the system. This limitation is discussed later in the paper.

D. Evaluation Metrics

The evaluation reports Recall@1, Recall@5, Recall@10, mean reciprocal rank (MRR), mAP@10, median rank, and false-positive rate for the negative controls.

For each query q in the query set Q, let r_q represent the rank of the first relevant result. Recall@K is defined as

$$R@K = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}(r_q \leq K). \tag{7}$$

MRR averages the reciprocal rank of the first relevant result:

$$\mathrm{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \begin{cases} 1/r_q, & r_q < \infty, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

mAP@10 represents the mean average precision over the ten highest-ranked results. Median rank is the middle value of the first relevant-result ranks.

For negative controls, the false-positive rate (FPR) represents the percentage of negative queries for which a positive match was accepted. Latency is reported using the mean, median, and 95th percentile (P95), measured from query submission to the final top-K list.

VI. RESULTS AND DISCUSSION

A. Retrieval Performance

The results show a steady improvement as additional capabilities are added from B1 to B5. R@1 increases from 52.3% to 68.2%, while R@10 increases from 80.4% to 91.6%. MRR moves from 0.61 to 0.75 and mAP@10 increases from 55.8% to 73.8%. The median rank of the first relevant result decreases from 4 to 2, while the negative-control FPR decreases from 21.0% to 8.0%.

TABLE III
RETRIEVAL PERFORMANCE ACROSS CUMULATIVE CONFIGURATIONS

Config.	R@1	R@5	R@10	MRR	mAP@10	Med. Rank	FPR
B1	52.3	70.1	80.4	0.61	55.8	4	21.0
B2	57.9	75.7	84.1	0.65	61.4	3	17.0
B3	62.6	79.4	87.9	0.69	66.8	3	13.0
B4	65.4	82.2	89.7	0.72	70.5	2	11.0
B5	68.2	84.1	91.6	0.75	73.8	2	8.0

R@1, R@5, R@10, mAP@10, and FPR are percentages. Med. Rank denotes the median rank of the first relevant result.

Each configuration adds a specific capability. B2 adds object evidence and track association, B3 adds attribute verification, B4 introduces relational reasoning, and B5 completes the multimodal pipeline with speech, OCR, and evidence linking.

Across the complete progression, R@1 improves by 15.9 percentage points and mAP@10 improves by 18.0 points. At the same time, the negative-control FPR decreases by 13 percentage points.

B. Query Latency

Additional processing stages also increase query processing time, as shown in Table IV. Mean latency increases from 1.84 s for B1 to 2.81 s for B5. P95 latency increases from 2.71 s to 4.08 s.

The increase is mainly associated with object retrieval, attribute verification, and relational processing.

TABLE IV
QUERY LATENCY ACROSS CUMULATIVE CONFIGURATIONS

Configuration	Mean (s)	Median (s)	P95 (s)
B1: Frame-Semantic	1.84	1.62	2.71
B2: Object-Aware	2.03	1.78	2.96
B3: Attribute-Aware	2.31	2.05	3.34
B4: Relational	2.57	2.24	3.71
B5: Full ChronoLens	2.81	2.46	4.08

C. Discussion

Frame-level semantic retrieval provides a useful starting point, but the results show that structured evidence from objects, attributes, and relations changes both retrieval accuracy and ranking. The complete B5 configuration reaches 68.2% R@1, 91.6% R@10, 0.75 MRR, and 73.8% mAP@10, with a median rank of 2.

The additional processing comes with a measurable latency cost. Mean latency increases by approximately 0.97 s compared with B1, reflecting the extra retrieval and verification stages.

There are several limitations to the current evaluation. First, all queries were tested on one video, so the results describe the behaviour of the system on this source and do not establish performance across different scenes or camera conditions. Second, the ground truth was constructed with assistance from ChronoLens evidence, which means the evaluation is not fully independent. Independent annotation would provide a stronger measurement.

Third, some query categories are small. In particular, the benchmark contains only four speech queries and three OCR queries, so their results should be interpreted cautiously. Fourth, the hybrid-ranking weights were selected heuristically rather than through a formal optimization procedure. The observed gains also depend on the accuracy of detection, tracking, and attribute extraction. Occlusion, viewpoint changes, and low visibility of attributes can reduce this accuracy.

Finally, appearance-based searching for people in surveillance footage can be misused. Practical deployments should therefore include appropriate access control, logging, and a lawful basis for using the system.

VII. CONCLUSION

ChronoLens is a multimodal video retrieval framework that combines frame-level semantic search with object, attribute, and spatial-temporal relational checks. The resulting evidence remains connected to persistent tracks and stored metadata, allowing retrieved moments to be traced back to the original video.

Using a fixed 107-query benchmark, performance improved across the five cumulative configurations. The complete B5 system achieved 68.2% R@1, 84.1% R@5, 91.6% R@10, 0.75 MRR, and 73.8% mAP@10, while the negative-control FPR decreased to 8.0%.

The results indicate that structured multimodal evidence can provide useful information beyond frame-level semantic similarity. However, the current evaluation uses a single video and ground truth that was partly derived from the system's own evidence. Therefore, further evaluation on larger and more varied video collections is needed.

Future work will focus on larger datasets, independently verified annotations, and reducing the computational cost of retrieval and relational reasoning.

REFERENCES

[1] Y. Tevissen, K. Guetari, and F. Petitpont, "Towards retrieval augmented generation over large video libraries," in *Proc. 16th Int. Conf. Human System Interaction (HSI)*, Paris, France, 2024, arXiv:2406.14938.

[2] K. Dela Rosa, "Video enriched retrieval augmented generation using aligned video captions," arXiv:2405.17706, 2024.

[3] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 8748–8763.

[4] H. Luo *et al.*, "CLIP4Clip: An empirical study of CLIP for end-to-end video clip retrieval and captioning," *Neurocomputing*, vol. 508, pp. 293–304, 2022.

[5] J. Lei, T. L. Berg, and M. Bansal, "Detecting moments and highlights in videos via natural language queries," in *Proc. Adv. Neural Information Processing Systems (NeurIPS)*, 2021.

[6] Y. Zhang *et al.*, "ByteTrack: Multi-object tracking by associating every detection box," in *Proc. Eur. Conf. Computer Vision (ECCV)*, 2022.

[7] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE Int. Conf. Image Processing (ICIP)*, 2017, pp. 3645–3649.

[8] A. Radford *et al.*, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023.

[9] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[10] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.

[11] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, "Ultralytics YOLO26: Unified real-time end-to-end vision models," arXiv:2606.03748, 2026.

[12] S. Li, T. Xiao, H. Li, B. Zhou, D. Yue, and X. Wang, "Person search with natural language description," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 1970–1979.

[13] M. Ye, J. Shen, G. Lin, T. Xiang, L. Shao, and S. C. H. Hoi, "Deep learning for person re-identification: A survey and outlook," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 6, pp. 2872–2893, Jun. 2022.

[14] M. Cherti *et al.*, "Reproducible scaling laws for contrastive language-image learning," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 2818–2829.

[15] C. Du *et al.*, "PP-OCR: A practical ultra lightweight OCR system," arXiv:2009.09941, 2020.